

AmplifiedCoaster: Virtual Roller Coaster Experience using Motorized Ramps and Personal Mobility Vehicle

Shunta Ito

shunta_ito@keio.jp

Graduate School of Media and Governance,
Keio University
Fujisawa, Japan

Yasuto Nakanishi

naka@keio.jp

Faculty of Environment and Information Studies,
Keio University
Fujisawa, Japan

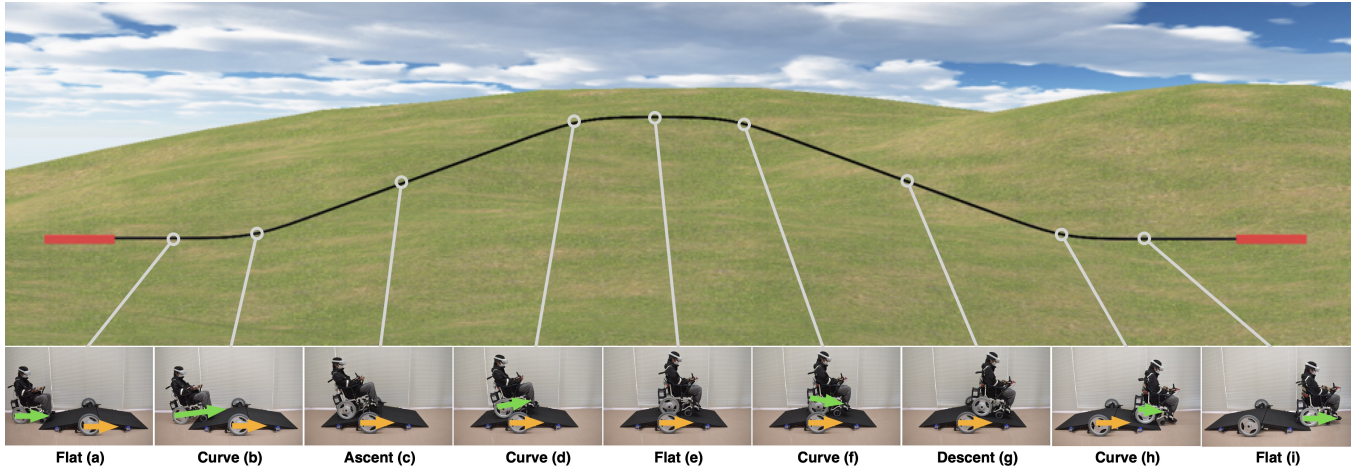


Figure 1: Schematic diagram of the ascent and descent experience using our system

ABSTRACT

A novel virtual reality (VR) ride consisting of a head-mounted display (HMD) and an electric wheelchair was developed for a virtual roller coaster experience. The VR ride was integrated with an electric wheeled ramp to amplify the perception of virtual ascent and descent. The ascending and descending motion on a slope was replicated continuously with varying curvatures. The system features an electric wheelchair and two motorized ramps with a fixed 10-degree angle, which the wheelchair can drive on and off to simulate ascending and descending in VR. Moreover, by adjusting the relative speed and position of the wheelchair and ramps, we can simulate pitch angle curvature in VR. We are investigating several methods for providing multiple ascent and descent experiences, and introduce one of them.

CCS CONCEPTS

• Human-centered computing → Human computer interaction (HCI).

KEYWORDS

Virtual Reality, Redirected Walking, Personal Mobility Vehicle, Electric Wheelchair, Motion Platform, Attraction

ACM Reference Format:

Shunta Ito and Yasuto Nakanishi. 2023. AmplifiedCoaster: Virtual Roller Coaster Experience using Motorized Ramps and Personal Mobility Vehicle. In *The 36th Annual ACM Symposium on User Interface Software and Technology (UIST '23 Adjunct)*, October 29–November 01, 2023, San Francisco, CA, USA. ACM, New York, NY, USA, 3 pages. <https://doi.org/10.1145/3586182.3616662>

1 INTRODUCTION

Attractions in amusement parks are excellent entertainment systems that provide extraordinary experiences by presenting users with various motion feedbacks through physical movements. Motion platforms (MPs) [5, 7] have been studied to provide such realistic experiences by combining visual content through an HMD or a large display. In recent years, researchers have shown increased interest in using personal mobility vehicles such as electric wheelchairs or cars for virtual reality (VR) attractions[1–3, 6].

To realize a virtual roller coaster experience, we have also developed a novel system that combines a VR ride composed of an electric wheelchair and an HMD with an electric-wheeled ramp [4]. In this system, the electric wheelchair rides up and down the electric-wheeled ramp to present the sensation of ascending and descending to the user, and the visual stimuli from the HMD amplify these sensations, providing a sense of ascending and descending, as in a roller coaster. Thus, we named it "AmplifiedCoaster."

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

UIST '23 Adjunct, October 29–November 01, 2023, San Francisco, CA, USA

© 2023 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-0096-5/23/10.

<https://doi.org/10.1145/3586182.3616662>

2 SYSTEM OVERVIEW

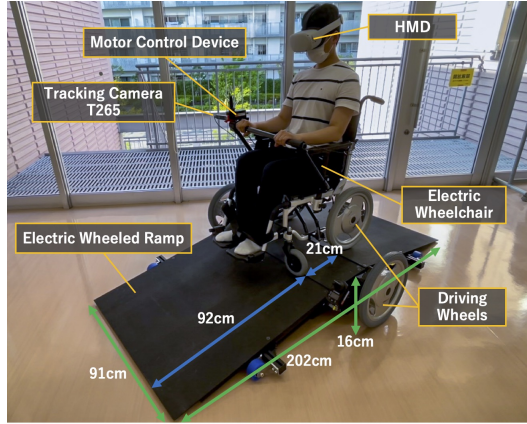


Figure 2: Hardware of Our System: Motorized Ramp, Personal Mobility Vehicle and HMD.

The hardware configuration of our system comprises an electric wheelchair (Yamaha TOWNYJOY X PLUS+) and an electric-wheeled ramp utilizing the same drive wheels (Fig.2). The electric-wheeled ramp is trapezoidal in shape, with dimensions of 91 cm width, 202 cm depth, and a maximum height of 16 cm. It consists of two inclined surfaces and one flat surface, measuring 92 cm and 21 cm in length, respectively. For safety reasons, it is recommended that the electric wheelchair moves on a ramp with a maximum inclination of 10°. Therefore, we set the angle of the electric-wheeled ramp to 10°. Additionally, a grab bar is installed in front of the electric wheelchair to ensure user safety, and the ramp's surface is coated with non-slip paint to prevent wheelchair slippage.

The motor control of both the electric wheelchair and the electric-wheeled ramp is achieved through a motor-control device provided by Yamaha Motor Co., Ltd.. This motor control device facilitates the transmission and reception of motor speed data (rotations per minute [rpm]) via serial communication with a single-board computer (Raspberry Pi 4 Model B). To capture the orientation data of the wheelchair, an Intel Realsense Tracking Camera T265 is integrated into the system.

To simulate the VR roller coaster experience, participants utilized the Meta Quest 2 headset (horizontal field of view 110°, refresh rate 120Hz during Oculus Link, resolution 1832 × 1920). The headset enabled real-time communication (Oculus Link) with the main PC. Visual scenes were generated using the Unity 2020.3.16 f1 game engine. The audio of the roller coaster was played through the built-in speakers of the headset.

Robot Operating System 2 (ROS 2) was utilized to transmit motor velocity (rpm) to the wheelchair and electric-wheeled ramp, as well as to receive orientation data from the wheelchair. The integration of ROS2 and Unity was achieved using the Unity-Robotics-Hub.

The velocity of the ramp was set to 0.28 m/s (1 km/h), as determined in a preliminary experiment where it was found to be the minimum speed required to evoke a sensation of movement. If the velocity in the VR space matches that of the real space, the travel distance in the VR roller coaster is limited by the physical

distance. A previous study [8] showed that an electric wheelchair could drive at higher speeds in the VR space compared to the real world without inducing significant discomfort. Therefore, the VR coaster drove at 1.5 m/s, which is approximately 5.4 times the speed of the real-world.

3 MULTIPLE ASCENTS AND DESCENTS EXPERIENCE

Our system of placing an electric wheelchair on a moving ramp can present the feeling of continuous ascent and descent. Furthermore, the electric wheelchair rides on and off a moving ramp, offering the user the perception of an ascending and descending slope with a varying curvature that combines translation and pitch directional rotation. This system has the advantage of providing an experience that combines vertical rotational motion and continuous operation among the virtual roller coaster experience (Fig.1).

In the transition from horizontal translation (Fig.1(a)) to ascent, the center axis of the front wheels of the wheelchair grounds at the edge of the ramp's surface and begins to ascend (Fig.1(b)). In presenting the ascent experience, the wheelchair is placed on the inclined surface of the ramp, and only the ramp moves (Fig.1(c)). In the transition from ascent to horizontal translation, the wheelchair ascends from the inclined surface to the flat surface of the ramp (Fig.1(d, e)). After the transition to horizontal translation, the wheelchair stops on the ramp, and only the ramp moves (Fig.1(e)). Then the wheelchair descended from the flat surface to another inclined surface of the ramp (Fig.1(f)). When presenting the descent experience, the wheelchair is placed on the inclined surface of the ramp, and only the ramp moved (Fig.1(g)). In descending on a curved rail, the electric wheelchair moves forward to the edge of the inclined surface of the ramp, and then gets off the slope (Fig.1(h)), and moves on the ground (Fig.1(i)).

By using multiple electric-wheeled ramps and presenting the ascent and descent sequentially, multiple ascent and descent experiences can be presented. As the wheelchair ascends and descends on a ramp, this method will provide a highly realistic experience. However, trapezoidal moving ramps must be placed for the number of ascents and descents, or to dynamically reposition the ramps as the wheelchair travels, which increases the cost of the apparatus and control.

4 CONCLUSION AND FUTURE WORK

This paper introduced our system that provides a virtual roller coaster experience; however, it is still only part of the roller coaster experience; curves to the side or accelerated descent have not yet been realized. We currently operate the ramp at the minimum speed necessary to present the stimuli, although it can reach speeds of up to 6 km/h. The ramp is a kind of differential two-wheeled robot controlled by ROS 2, which ideally should enable it to accelerate and decelerate while navigating complex paths. Our next goal is to create an experience of accelerating descent and ascent, combining more complex physical and visual stimuli.

ACKNOWLEDGMENTS

This work was supported by JST, CREST Grant Number JPMJCR19A4, Japan.

REFERENCES

- [1] Hyung-Jun Cho and Gerard J. Kim. 2022. RideVR: Reducing Sickness for In-Car Virtual Reality by Mixed-in Presentation of Motion Flow Information. *IEEE Access* 10 (2022), 34003–34011. <https://doi.org/10.1109/ACCESS.2022.3162221>
- [2] John Paulin Hansen, Astrid Kofod Trudslev, Sara Amdt Harild, Alexandre Alapetite, and Katsumi Minakata. 2019. Providing Access to VR Through a Wheelchair. *Extended Abstracts of the 2019 CHI Conference on Human Factors in Computing Systems* (2019), 1–8. <https://doi.org/10.1145/3290607.3299048>
- [3] Philipp Hock, Mark Colley, Ali Askari, Tobias Wagner, Martin Baumann, and Enrico Rukzio. 2022. Introducing VAMPIRE – Using Kinaesthetic Feedback in Virtual Reality for Automated Driving Experiments. In *Proceedings of the 14th International Conference on Automotive User Interfaces and Interactive Vehicular Applications* (Seoul, Republic of Korea) (*AutomotiveUI '22*). Association for Computing Machinery, New York, NY, USA, 204–214. <https://doi.org/10.1145/3543174.3545252>
- [4] Shunta Ito and Yasuto Nakanishi. 2022. AmplifiedCoaster: Amplifying the Perception of Ascent and Descent in Virtual-Reality-Equipped Electric Wheelchair in an Electric Wheeled Ramp. In *ICAT-EGVE 2022 - International Conference on Artificial Reality and Telexistence and Eurographics Symposium on Virtual Environments - Posters and Demos*, Theophilus Teo and Ryota Kondo (Eds.). The Eurographics Association. <https://doi.org/10.2312/egve.20221312>
- [5] Philipp Miermeister, Maria Lächele, Rainer Boss, Carlo Masone, Christian Schenk, Joachim Tesch, Michael Kerger, Harald Teufel, Andreas Pott, and Heinrich H. Bühlhoff. 2016. The CableRobot simulator large scale motion platform based on cable robot technology. In *2016 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. 3024–3029. <https://doi.org/10.1109/IROS.2016.7759468>
- [6] Jan Gugenheimer Philipp Hock, Sebastian Benedikter and Enrico Rukzio. 2017. CarVR: Enabling In-Car Virtual Reality Entertainment. In *Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems*. ACM, 4034–4044. <https://doi.org/10.1145/3025453.3025665>
- [7] D. Stewart. 1965. A Platform with Six Degrees of Freedom. *Proceedings of the Institution of Mechanical Engineers* 180, 1 (1965), 371–386. https://doi.org/10.1243/PIME_PROC_1965_180_029_02
- [8] Sakai Yusuke, Watanabe Toshimitsu, Ishiguro Yoshio, Nishino Takanori, and Takeda Kazuya. 2019. Effects on User Perception of a 'modified' Speed Experience through in-Vehicle Virtual Reality. In *Proceedings of the 11th International Conference on Automotive User Interfaces and Interactive Vehicular Applications: Adjunct Proceedings*. ACM, New York, NY, USA, 166–170. <https://doi.org/10.1145/3349263.3351335>